1 **(a)** 0.17 **[B1]**

(b) The train does not arrive on time (or equivalent) **[B1]**

[Total Marks: 2]

2 **(a)** $1 - \left(\frac{2}{7} + \frac{1}{3}\right)$ or equivalent **[M1]**

$\frac{8}{21}$ **[A1]**

(b) $\frac{5}{7}$ **[B1]**

[Total Marks: 3]

3 **(a)** $0.15 + 2x + 0.31 + x + 0.24 = 1$ or $3x + 0.7 = 1$ **[M1]**

0.1 **[A1]**

(b) $P(2) = 2(0.1) = 0.2$ **[M1]**

0.8 **[A1]**

[Total Marks: 4]

4 **(a)** Sum of given probabilities: $0.15 + 0.3 + 0.45 = 0.9$ **[M1]**

0.1 **[A1]**

(b) 0.45 **[B1]**

(c) Uses $1 - P(G)$ formula: $1 - 0.45$ **[M1]**

0.55 **[A1]**

[Total Marks: 5]

5 **(a)** $0.15 + 2x + 0.35 + 3x = 1$ **[M1]**

$5x = 0.50$, so $x = 0.1$ (or 0.10) **[A1]**

(b) **(i)** $P(A) = P(2) + P(4) = 0.20 + 0.30$ **[M1]**

0.5 (or 0.50 or $\frac{1}{2}$) **[A1]**

(ii) $1 - 0.5 = 0.5$ (or 0.50 or $\frac{1}{2}$) **[B1]**

(c) $P(2) + P(3) = 0.20 + 0.35 = 0.55$ **[M1]**

$200 \times 0.55 = 110$ **[A1]**

[Total Marks: 7]

6 **(a)** $\frac{1}{5} + 0.35 + x + 2x = 1$ or $0.55 + 3x = 1$ **[M1]**

0.15 **[A1]**

(b) $1 - 2(0.15)$ or $\frac{1}{5} + 0.35 + 0.15$ **[M1]**

0.7 or $\frac{7}{10}$ **[A1]**

[Total Marks: 4]

7 **(a)** $p^2 + p + 0.15 + 0.1 = 1$ **[M1]**

$p^2 + p - 0.75 = 0$ leading to $4p^2 + 4p - 3 = 0$ with no errors seen **[A1]**

(b) $(2p - 1)(2p + 3) = 0$ or correct substitution into the quadratic formula **[M1]**

0.5 (or $\frac{1}{2}$) **[A1]**

(c) $P(E) = 0.5 + 0.1 = 0.6$ or $P(E') = P(1) + P(3)$ **[M1]**

0.4 (or $\frac{2}{5}$) **[A1]**

[Total Marks: 6]

8 **(a)** Places $2x$ in the intersection **[B1]**

Places $38 - 2x$ in T only region and $24 - 2x$ in C only region **[B1]**

Places x in the region outside both circles **[B1]**

(b) Forms correct equation: $(38 - 2x) + 2x + (24 - 2x) + x = 60$ or $62 - x = 60$ **[M1]**

2 **[A1]**

(c) Finds the number of people in $T \cap C'$: $38 - 2(2) = 34$ **[M1]**

$\frac{34}{60}$ or $\frac{17}{30}$ **[A1]**

[Total Marks: 7]

9 **(a)** $x^2 + 3x + 2$ **[B1]**

(b) $\frac{x^2}{x^2 + 3x + 2} = \frac{8}{15}$ **[M1]**

$15x^2 = 8(x^2 + 3x + 2) \implies 15x^2 = 8x^2 + 24x + 16$ **[M1]**

(c) $(7x + 4)(x - 4) = 0$ or correct substitution into the quadratic formula [M2]
(Award M1 for $(7x + a)(x + b) = 0$ where $ab = -16$ or $7b + a = -24$)

4 **[A1]**

(d) Total counters $= 4^2 + 3(4) + 2 = 30$ and number of blue counters $= 3(4) = 12$ **[M1]**

$\frac{18}{30} = \frac{3}{5}$ **[A1]**

[Total Marks: 8]

10 **(a)** Total number of counters $= x + (2x - 1) + (x^2 - 4) = x^2 + 3x - 5$ **[B1]**

$\dfrac{x^2 - 4}{x^2 + 3x - 5} = \dfrac{3}{5}$ giving $5(x^2 - 4) = 3(x^2 + 3x - 5)$ **[M1]**

$5x^2 - 20 = 3x^2 + 9x - 15$ simplifying to $2x^2 - 9x - 5 = 0$ **[A1]**

(b) $(2x + 1)(x - 5) = 0$ **[M1]**

5 (rejecting $x = -\dfrac{1}{2}$ since $x > 2$) **[A1]**

(c) **(i)** Total counters $= 5 + 9 + 21 = 35$, giving $P(R) = \dfrac{5}{35}$ **[M1]**

$\dfrac{1}{7}$ **[A1]**

(ii) $P(G') = 1 - P(G) = 1 - \dfrac{3}{5}$ **[M1]**

$\dfrac{2}{5}$ **[A1]**

(d) Number of counters that are not red $= 9 + 21 = 30$, giving $\dfrac{30}{35 + k} = \dfrac{3}{4}$ **[M1]**

$3(35 + k) = 120$ giving $35 + k = 40$ **[M1]**

5 **[A1]**

[Total Marks: 12]

11 **(a)** $1 - \left(\dfrac{2}{5} + \dfrac{1}{3}\right)$ or $1 - \dfrac{11}{15}$ **[M1]**

$\dfrac{4}{15}$ **[A1]**

(b) Equates their fraction to the number of green marbles: $\dfrac{4}{15}$ of total $= 12$ **[M1]**

Total $= 12 \times \dfrac{15}{4}$ **[M1]**

45 **[A1]**

(c) $\dfrac{2}{3}$ **[B1]**

[Total Marks: 6]

12 **(a)** $14 + x + 9 + 5 = 36$ or equivalent **[M1]**

8 **[A1]**

(b) $\dfrac{14 + 8}{36}$ or $\dfrac{22}{36}$ **[M1]**

$\dfrac{11}{18}$ **[A1]**

(c) $1 - P(B)$ or $\dfrac{14 + 5}{36}$ **[M1]**

$\dfrac{19}{36}$ **[A1]**

[Total Marks: 6]

13 **(a)** 21 in the intersection $C \cap D$ **[B1]**

34 in C only and 47 in D only **[B1]**

18 outside both circles **[B1]**

(b) $\frac{21}{120}$ or $\frac{7}{40}$ **[B1]**

(c) $1 - \frac{68}{120}$ or $\frac{34+18}{120}$ **[M1]**

$\frac{52}{120}$ or $\frac{13}{30}$ **[A1]**

(d) $\frac{34+47}{120}$ or $\frac{120-21-18}{120}$ **[M1]**

$\frac{81}{120}$ or $\frac{27}{40}$ **[A1]**

[Total Marks: 8]

14 **(a)** 36 **[B1]**

(b) **(i)** $n(M) = 38 + 17 = 55$, giving $\frac{55}{120}$ **[M1]**

$\frac{11}{24}$ **[A1]**

(ii) $n(M' \cap D) = 29$ **[B1]**

$\frac{29}{120}$ **[B1]**

(iii) $1 - \frac{17}{120}$ (or $\frac{38+29+36}{120}$) **[M1]**

$\frac{103}{120}$ **[A1]**

(c) Total studying Music is 55, and 17 also study Drama **[M1]**

$\frac{17}{55}$ **[A1]**

[Total Marks: 9]

15 **(a)** $(x+4) + x + (2x-1) + (3x+5) = 50$ or $7x + 8 = 50$ **[M1]**

6 **[A1]**

(b) $(2(6)-1) + (3(6)+5)$ or $50 - (6+4+6)$ or $\frac{11+23}{50}$ **[M1]**

$\frac{34}{50}$ or $\frac{17}{25}$ or 0.68 **[A1]**

(c) $(6+4) + (2(6)-1)$ or $10 + 11$ or $\frac{21}{50}$ **[M1]**

$\frac{21}{50}$ or 0.42 **[A1]**

[Total Marks: 6]